KENDRIYA VIDYALAYA SANGATHAN BHOPAL REGION

PRE-BOARD I SET 2

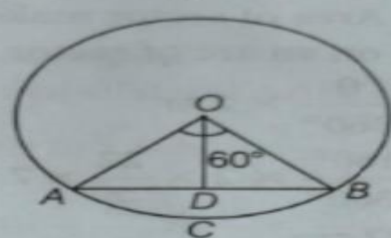
MARKING SCHEME SESSION 2024-25

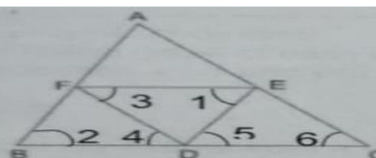
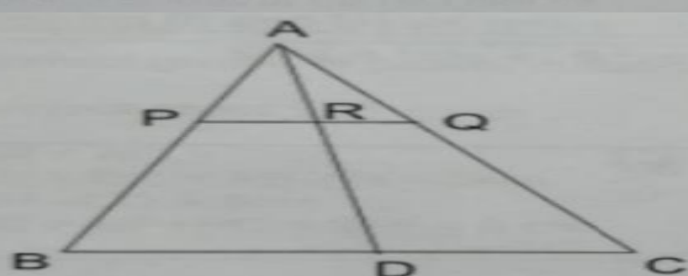
CLASS : - X SUBJECT :- MATHEMATICS(STANDARD)

TIME :- 3 HOURS MAX.MARKS: 80

Q.NO.	ANSWER	MARKS
	SECTION :A	
1.	(d) $\frac{2\pi}{3}$ cm^3	1
2.	(b) 2500	1
3.	(b) 1/5	1
4.	(d) coincident	1
5.	(c) 5	1
6.	(a) $x^2-4x+1=0$	1
7.	(d) 125^0	1
8.	(c) 13	1
9.	(c) $ac = \frac{b^2}{4}$	1
10.	(d) 40^0	1
11.	(c) 1/m	1
12.	(c) 7	1
13.	(a) 82	1
14.	(b) 462 cm^2	1
15.	(a) (0, - 3)	1
16.	(c) 45^0	1
17.	(b) 12 cm	1
18.	(d) IV quadrant	1
19.	(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)	1
20.	(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)	1
	SECTION B	
21	$(10)^2 = (9 - x)^2 + (8 - 2)^2$ $100 = 81 - 18x + x^2 + 36$ $x^2 - 18x + 17 = 0$	½ ½ ½

	$x^2 - 17x - x + 17 = 0$ $x(x - 17) - 1(x - 17) = 0$ $x = 17$, $x = 1$	 ½
22	Let the ratio be k : 1 and point on X-axis is (x, 0) $0 = \frac{k(-5) + 1(3)}{k+1}$ $-5k + 3 = 0$ $k = \frac{3}{5}$ so ratio is 3: 5 OR For perimeter of triangle find the lengths of sides Distance $AB^2 = (0 - 1)^2 + (0 - 0)^2$ $BC^2 = (1 - 0)^0 + (0 - 1)^2$ $AC^2 = (0 - 0)^2 + (1 - 0)^2$ So perimeter = $1 + 1 + \sqrt{2} = 2 + \sqrt{2}$	½ 1 ½ ½ ½ ½ ½
23	LCM of 48, 72 and 108 = 2 ×2 × 2× 2 × 3 × 3 × 3 = 432 Therefore, after 432 seconds, they will change simultaneously. We know that 60 sec. = 1 min. Therefore 432 sec. = $\frac{432}{60}$ = 7 min. 12 sec. Hence, the lights change simultaneously at 7:07:12 am	 1 ½ ½
24	$2 \times (2)^2 + x \times \left(\frac{\sqrt{3}}{2}\right)^2 - \frac{3}{4} \times \left(\frac{1}{\sqrt{3}}\right)^2 = 10$ $8 + \frac{3x}{4} - \frac{\cancel{3}}{4} \times \frac{1}{\cancel{3}} = 10$ $8 - \frac{1}{4} + 3x = 10$ $3x = 10 - 8 + \frac{1}{4}$ $3x = \frac{2}{1} + \frac{1}{4}$ $3x = \frac{8+1}{4}$ $3x = \frac{9}{4}$ $x = \frac{\cancel{9}\,3}{4 \times \cancel{3}}$ $\boxed{x = \frac{3}{4}}$	1 ½ ½
25	Let the angles be a^0, $(a + d)^0$, $(a + 2d)^0$ Then, we get a + a + d + a + 2d = 180 and a + 2d = 2a after solving angles will be 40^0, 60^0, 80^0 OR We know that 18 is the first multiple of 9 lies between 10 and 300 and last multiple of 9 up to 300 is 297. $a_n = a + (n - 1)d$ $297 = 18 + (n - 1)9$	½ ½ ½ ½ ½

	$297 - 18 = (n-1)9$ $n = 32$	½ ½ ½
	SECTION C	
26	Stepwise marks for correct proof	3
27	LHS = $\frac{cos^2 A + (1+\sin A)^2}{\cos A(1+\sin A)}$ LHS = $\frac{cos^2 A + sin^2 A + 1 + 2\sin A}{(1+\sin A)\cos A}$ LHS = $\frac{1+1+2\sin A}{(1+\sin A)\cos A}$ LHS = $\frac{2(1+sinA)}{(1+\sin A)\cos A}$ LHS = $\frac{2}{\cos A}$ = 2sec A OR STEP WISE MARKS FOR CORRECT SOLUTION	½ 1 ½ ½ ½
28	Chords AB subtends an angle 60° at the centre.  $\therefore \quad \angle AOB = 60°$ and $\angle OAB = \angle OBA = x$ [say] [∵ angles opposite to equal sides are equal] In ΔAOB, $\angle AOB + \angle OBA + \angle OAB = 180°$ $\Rightarrow 60° + x + x = 180° \Rightarrow x = 60°$ So, ΔAOB is an equilateral triangle. Now,area of $\Delta AOB = \frac{\sqrt{3}}{4}a^2 = \frac{\sqrt{3}}{4} \times (21)^2 = 190.95\ cm^2$ Area of sector $OACBD = \frac{\theta}{360°} \times \pi r^2$ $= \frac{60°}{360°} \times 3.14 \times (21)^2 = 230.79\ cm^2$ $\therefore$ Area of minor segment $ACBDA$ = Area of sector $OACBO$ – Area of ΔAOB $= (230.79 - 190.95)\ cm^2 = 39.84\ cm^2$	1 1 1

29	For getting $\alpha + \beta = 1 \; and \; \alpha\beta = 1/5$ For getting $\alpha^2 + \beta^2 = \frac{4}{5}$ For getting $\alpha^{-1} + \beta^{-1} = \frac{1}{5}$	1 1 1
30	Let one number be x. Then, another number = (11- x) According to question, $\frac{1}{x} + \frac{1}{11-x} = \frac{11}{28}$ $\frac{11-x+x}{x(11-x)} = \frac{11}{28}$ $\frac{11}{x(11-x)} = \frac{11}{28}$ $x(11 - x) = 28$ $x^2 - 11x + 28 = 0$ $x = 4, x = 7$ so two numbers are 4 and 7.	
31	 Since D, E, F are the mid points of BC, CA, AB respectively Therefore, EF||BC, DF||AC, DE||AB BDEF is a parallelogram $\angle 1 = \angle 2$ & $\angle 3 = \angle 4$ $\Delta FBD \sim \Delta DEF$ Also, DCEF is a parallelogram $\angle 3 = \angle 6$ & $\angle 1 = \angle 2$ (proved above) $\Delta DEF \sim \Delta ABC$  Since PQ//BC therefore $\Delta APR \sim \Delta ABD$ $\Rightarrow \frac{AP}{AB} = \frac{PR}{BD}$ .... (i) $\Delta AQR \sim \Delta ACD$ $\Rightarrow \frac{AQ}{AC} = \frac{RQ}{DC}$ .... (ii) Now, $\frac{AP}{AB} = \frac{AQ}{AC}$ ....(iii) Using (i), (ii) & (iii), $\frac{PR}{BD} = \frac{RQ}{DC}$ But, BD = DC $\Rightarrow$ PR = RQ or AD bisects PQ	1 1 1 OR 1 1 1
	SECTION D	

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Class (C.I.)	Frequency (f_i)	Class Mark (x_i)	$X_i f_i$	cf
40-45	8	42.5	340	8
45-50	9	47.5	427.5	17
50-55	10	52.5	525	27
55-60	9	57.5	517.5	36
60-65	5	62.5	312.5	41
65-70	4	67.5	270	45

Correct table 2marks

$\sum f_i = 45$ $\qquad\qquad$ $\sum f_i x_i = 2392.5$

Mean = $\frac{\sum f_i x_i}{\sum f_i} = \frac{2392.5}{45}$ = 53.167 — 1

Size of (N/2) th item is 45/2 ie. 22.5

Which lies in the class interval 50-55 — ½

Here, l = 50, h = 5, f = 10, cf = 17

Median = $l + \frac{\frac{N}{2} - cf}{f} \times h$ — ½

$= 50 + \frac{22.5-17}{10} \times 5 = 52.75$ — 1

Therefore, Mean = 53.167 and Median is 52.75

OR

Monthly Expenditure	f_i	x_i	$f_i\ x_i$
1000-1500	24	1250	30000
1500-2000	40	1750	70000
2000-2500	33	2250	74250
2500-3000	x	2750	77000
3000-3500	30	3250	97500
3500-4000	22	3750	82500
4000-4500	16	4250	68000
4500-5000	7	4750	33250

Correct table 2marks

172 + x = 200 — 1

x = 28 — ½

Mean = $\frac{532500}{200}$ — 1

= 2662.5 — ½

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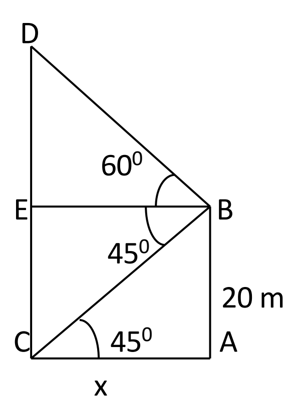
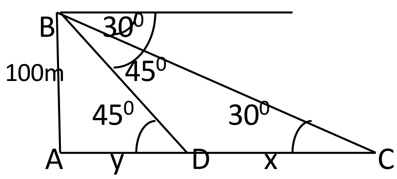
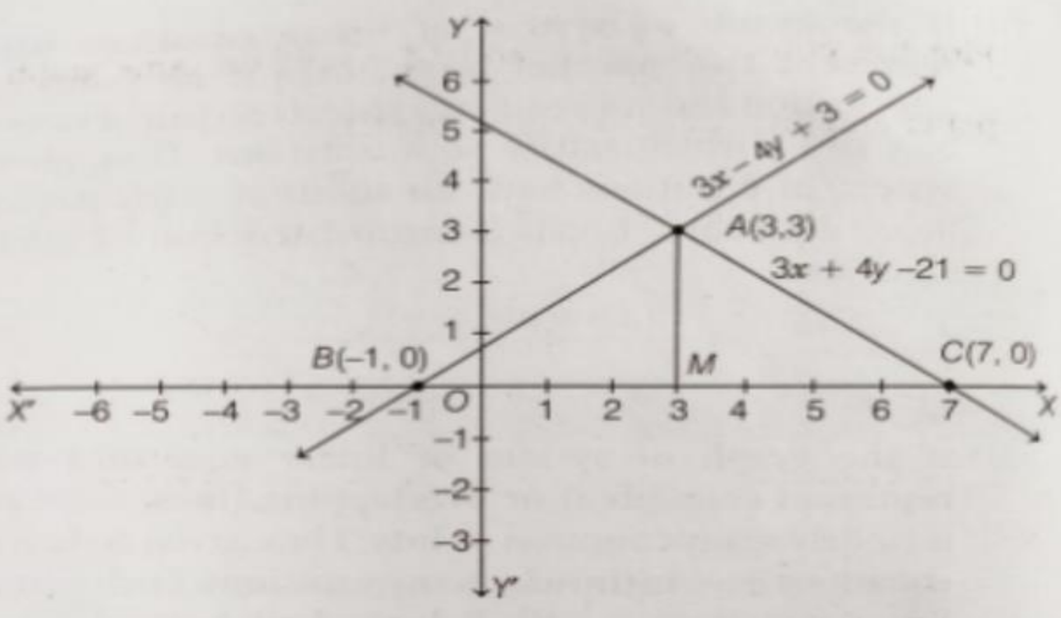


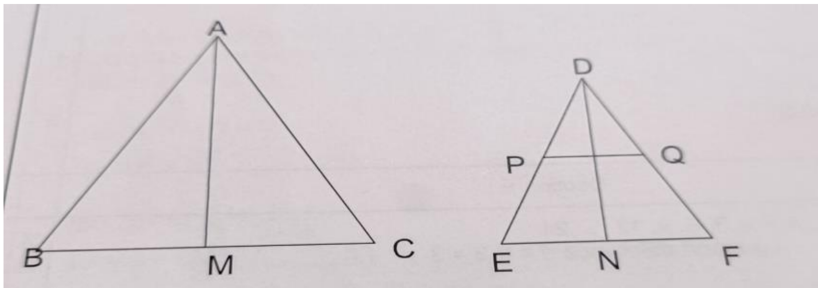
1 mark for correct figure

Let AB be the building and CD be the tower

	Here $\tan 60^0 = \sqrt{3} = \frac{h}{x}$ $h = x\sqrt{3}$ ....(i) $\tan 45^0 = \frac{20}{x} = 1$ x = 20 m ....(ii) (distance between tower and building) solving (i) and (ii) to get h = 20 × 1.732 = 34.64 m therefore, the height of the tower = h + 20 = 54.64m. OR  Let AB be the light house and C and D be the positions of ships. $\tan 30^0 = \frac{1}{\sqrt{3}} = \frac{100}{x+y}$ $x + y = 100\sqrt{3}$ ....(i) $\tan 45^0 = 1 = \frac{100}{y}$ y = 100 ....(ii) Solving (i) and (ii) to get $x = 100(\sqrt{3} - 1)$ x = 100 × .732 x = 73.2m distance between the ships is 73.2 m	1 ½ 1 ½ ½ ½ 1 mark for correct figure 1 ½ 1 ½ 1
34	Correct given, to prove, construction, figure Correct proof AR = AQ = 12 cm BP = BR = AB – AR = 3 cm CP = CQ = 7 cm BC = BP + PC = 3 + 7 = 10 cm	1 2 ½ ½ ½ ½

35	Given, pair of linear equations is $3x - 4y + 3 = 0$ ...(i) and $3x + 4y - 21 = 0$ ...(ii) Table for $3x - 4y + 3 = 0$ or $y = \frac{3x+3}{4}$ is | x | 3 | -1 | | $y = \frac{3x+3}{4}$ | 3 | 0 | | Points | $A(3, 3)$ | $B(-1, 0)$ | Plot the points $A(3, 3)$ and $B(-1, 0)$ on the graph paper and join them to get the straight line AB. Now, table for $3x + 4y - 21 = 0$ or $y = \frac{21-3x}{4}$ is | x | 3 | 7 | | $y = \frac{21-3x}{4}$ | 3 | 0 | | Points | $A(3, 3)$ | $C(7, 0)$ | Plot the points $A(3, 3)$ and $C(7, 0)$ on same graph paper and join them to get the straight line AC.  The lines AB and AC intersect each other at point $A(3, 3)$. So, the solution of the given pair of linear equations is $(3, 3)$.	Correct points of eqn 1, 1 mark Correct points of eqn 2, 1 mark Correct plotting on graph 2 marks Correct solution 1 mark.

	SECTION E	
36	Q36 (i) GOLF BALL $r = \frac{42}{2} = 2.1$ cm Dimple = Hemi sphere Radius $= r = 2$ mm $= 0.2$ cm S.A of dimple $= 2\pi r^2$ $= 2 \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10}$ $= \frac{176}{700} = 0.2514$ cm^2 — 01 (ii) vol. to make = vol. of Hemi sphere. $= \frac{2}{3}\pi r^3$ $= \frac{2}{3} \times \frac{22}{7} \times \frac{2}{10} \times \frac{2}{10} \times \frac{2}{10}$ $= 0.01676$ cm^3 — 01 = (iii) vol. of golf = vol. of sphere − vol of 315 Hemi sphere (iv) Ball $= \frac{4}{3}\pi r^3 - \frac{2}{3}\pi r^3 \times 315$ $= \frac{4}{3} \times \frac{22}{7} \times \frac{21}{10} \times \frac{21}{10} \times \frac{21}{10} - \frac{2}{3} \times \frac{22}{7} \times \left(\frac{2}{10}\right)^3 \times 315$ $= 38.808 - 5.28$ $= 33.528$ cm^3 — 02	4
37	(i)Number of students who do not prefer to walk = 400 – 180 = 220 P(selected student doesn't prefer to walk) $= \frac{220}{400}$ or $\frac{11}{20}$ (ii)Total number of students who prefer to walk or use bicycle = 180 + 100 = 280 P(selected student prefers to walk or use bicycle = 280/400 or 7/10 (iii)(A)50% of walking students who used bicycle = 90 Number of students who already use bicycle = 100 P(selected studentusesbicycle) = 190/400 or 19/40 OR (B)Number of students who preferred to be dropped by car = 400 – (180 + 100 + 40) P(selected student is dropped by car = 80/400 or 1/5	½ ½ ½ ½ ½ ½ 1 1 1
38 (i) (ii)	∠DPQ = ∠DEF ∠PDQ = ∠EDF Therefore ΔDPQ ~ΔDEF DE = 100 + 140 = 240 cm	 1 ½

(iii)(A)	$\frac{DP}{DE} = \frac{PQ}{EF}$ Therefore $\frac{PQ}{EF} = \frac{100}{240}$ or 5/12 $\frac{AB}{DE} = \frac{5}{2} = \frac{BC}{EF} = \frac{AC}{DF}$ AB = $\frac{5}{2}$ DE $\frac{perimeter\ of\ \Delta ABC}{perimeter\ of\ \Delta DEF} = \frac{\frac{5}{2}(DE+EF+FD)}{DE+EF+FD} = \frac{5}{2}$ (Constant) OR  $\frac{AB}{DE} = \frac{BC}{EF} = \frac{\frac{BC}{2}}{\frac{EF}{2}} \frac{BM}{EN}$ Also ∠B = ∠E Therefore ΔABM ~ΔDEN	½ 1 1 Correct fig. ½ 1 ½
	THE END	